Mars Habitat Automation Platform

Distributed Automation for a Simulated Mars Habitat

MLPG

Mars Operations – Microservices-based IoT Platform

Outline

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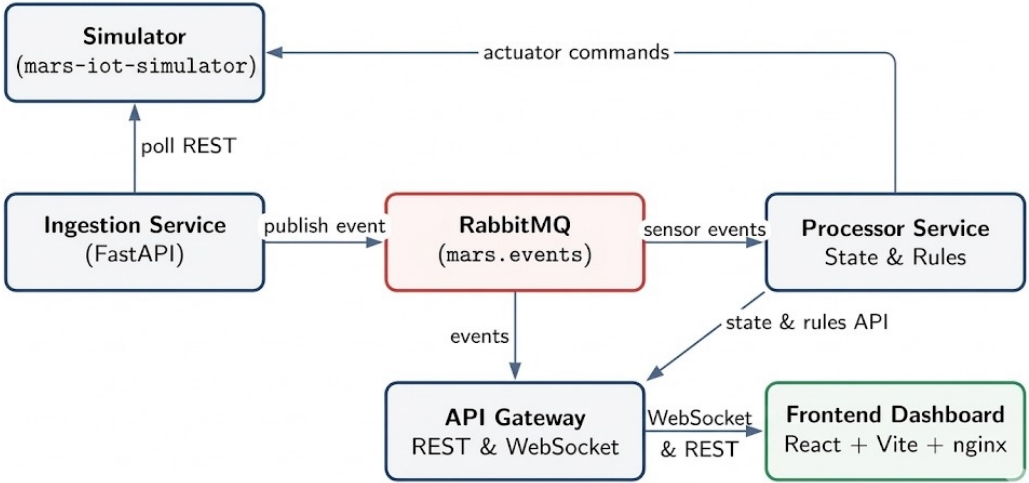
System Description

Goal

Provide an automation platform to **monitor** and **control** a simulated Mars habitat, starting from sensor data produced by the Mars IoT Simulator.

- **Unified data model**: all raw sensor and actuator signals are normalized into a single `UnifiedEvent` schema.
- **Event-driven architecture**: services exchange events through a RabbitMQ broker.
- **Automation rules engine**: IF–THEN rules evaluate incoming events and trigger actuator commands.
- **Manual control**: operators can set actuators ON/OFF from the dashboard.
- **Real-time dashboard**: a React SPA (Single Page Application) shows live readings, histories, rule status and conflicts.

High-level Data Flow



Docker Composition

- **simulator** (`port: 8080`): external Mars IoT Simulator, source of all sensor data and actuators.
- **rabbitmq** (`ports: 5672, 15672`): message broker, exchange `mars.events`, queues `processor.events`, `gateway.events`.
- **ingestion-service** (`port: 8001`): polls simulator REST (Representational State Transfer, a group of standard rules on how two programs should communicate, e.g. GET, POST, ...) endpoints and publishes normalized events.
- **processor-service** (`port: 8002`): state cache, rules engine, conflict arbitrator, SQLite persistence, actuator client.
- **api-gateway** (`port: 8003`): WebSocket endpoint `/ws`, REST proxies to processor and simulator.
- **frontend** (`port: 3000`): React SPA served via nginx (Engine-ex); proxies `/api` and `/ws` to the gateway.

Ingestion Service Responsibilities

Role in the platform

The Ingestion Service is the **entry point** for all sensor data coming from the Mars IoT Simulator.

- **Polling the simulator**
  - Periodically calls the simulator REST endpoints for each sensor (temperature, humidity, particulate, tank levels, ...).
- **Normalization into `UnifiedEvent`**
  - Converts heterogeneous payloads into the common `UnifiedEvent` schema.
- **Publishing to RabbitMQ**
  - Acts as a producer on the `mars.events` exchange.
  - It's fully stateless: in case of restart, it just reconnects and resumes polling the simulator.

Processor Service Responsibilities (1/2)

Core logic engine

The Processor Service is the **brain** of the platform: it tracks latest sensor state and decides when/how to act.

- **State management**
  - Consumes normalized events from RabbitMQ and updates in-memory `StateCache`.
  - Exposes REST endpoints for full/single-sensor state.
- **Rule evaluation**
  - Loads automation rules from SQLite and keeps a list of rules.
  - Evaluates conditions per incoming event and emits logical actuator commands.

Processor Service Responsibilities (2/2)

- **Conflict arbitration and safety**
  - Uses the `Arbitrator` to group commands per actuator in a short time window.
  - Detects ON/OFF conflicts and applies safe-state policy (default: `OFF`).
- **Persistence**
  - Stores rules in SQLite on a Docker volume so configuration survives restarts.

API Gateway and RabbitMQ

API Gateway

- Provides a **single entry point** for the frontend.
- Exposes `/ws` for live event streaming.
- Proxies `/api` calls to Processor (state/rules/conflicts) and simulator actuators.

RabbitMQ

- Central broker for sensor readings, actuator commands, and conflict alerts.
- Implements `mars.events` exchange with service queues.
- Decouples ingestion, processing, and streaming so services can scale/restart independently.

Frontend Dashboard Responsibilities

Operator-facing features

- Consumes WebSocket events and REST APIs from the gateway.
- Renders live sensor cards, actuator states, automation rules, and conflicts.
- Supports manual actuator control and rule actions (create/edit/toggle/delete).

Backend Technology Choices

Languages and frameworks

- **Python 3.11** for all internal services (ingestion, processor, API Gateway).
- **FastAPI** as the main web framework:
  - async-first, well aligned with message queues and WebSockets;

Supporting libraries (AIO versions nd Pydantic)

- **aio-pika**: asynchronous RabbitMQ client used by ingestion, processor and gateway to publish/consume events.
- **aiohttp**: async HTTP client to talk to the simulator and to proxy requests between services.
- **aiosqlite**: async SQLite driver used to store/retrieve rules without blocking the event loop.
- **Pydantic**: Ensures that the input data is correct.

Frontend and UX Technology Choices

Frontend stack

- **React 18** for building an interactive single-page dashboard capable of updating without refreshing the page.
- **Vite** as the build tool:
  - provides fast development pages

Serving and integration

- **nginx** serves the built static assets and proxies:
  - REST requests under `/api` to the API Gateway;
  - WebSocket connections under `/ws` to the same gateway.
- The frontend maintains its own in-memory state and histories, reacting to messages from the WebSocket and updates from REST.

Infrastructure Components

Message broker

- **RabbitMQ 3-management** is used as the central event bus.
- Defines the `mars.events` exchange and service-specific queues.

Database

- **SQLite** is chosen for persistence of automation rules:
  - embedded and lightweight; no external database service is required.
  - mounted on a Docker volume to survive container restarts.
- Only configuration data (rules) are persisted.

Complete List of User Stories (1/2)

1. **US01** – As an operator, I want to read sensors data, so that I can monitor the habitat.
2. **US02** – As an operator, I want to normalize sensors data, so that I can process them.
3. **US03** – As an operator, I want to create automation rules, so that I can automate the habitat.
4. **US04** – As an operator, I want to edit/delete existing automation rules, so that I can modify the behavior of the habitat.
5. **US05** – As an operator, I want to switch on/off automation rules, so that I can modify the behavior of the habitat.
6. **US06** – As an operator, I want to save automation rules, so that they persist after a restart.

Complete List of User Stories (2/2)

7. **US07** – As an operator, I want to interact manually with actuators, so that I can directly control the habitat.
8. **US08** – As an operator, I want to see the latest sensor readings, so that I can react immediately to critical changes in the habitat.
9. **US09** – As an operator, I want to check sensors status, so that I can detect malfunctions.
10. **US10** – As an operator, I want to see the evolution of sensors data, so that I can understand the behavior of the habitat.
11. **US11** – As the software, I want to show conflicts between automator rules, so that the operator can solve the issue.

User Stories (1/2): Monitoring and Data

- **US01 – US02**: **Read & Normalize** raw sensor data to enable continuous habitat monitoring.
- **US08**: **Live Readings** to react immediately to critical environmental changes.
- **US09**: **Sensor Status** to quickly detect anomalies and malfunctions.
- **US10**: **Data Evolution** to understand long-term habitat behavior through historical trends.

How we address them

Ingestion normalizes simulator data; the processor keeps the latest state; the dashboard shows live sensor cards, status badges and mini-charts.

User Stories (2/2): Automation, Control and Conflicts

- **US03 – US06**: **Rule Lifecycle Management** (create, edit, toggle, persist) to fully automate the habitat behavior.
- **US11**: **Rule Priority** to ensure critical automations take precedence.
- **US07**: **Manual Control** to directly interact with actuators.
- **US11**: **Conflict Detection** to automatically highlight overlapping rules so the operator can resolve them.

How we address them

Rules are stored and evaluated by the processor; the dashboard lets operators manage rules and control actuators; conflicts are strictly arbitrated and highlighted in the UI.

Conflict Resolution – Intuition

Context

Multiple automation rules can target the **same actuator** in the same evaluation cycle.

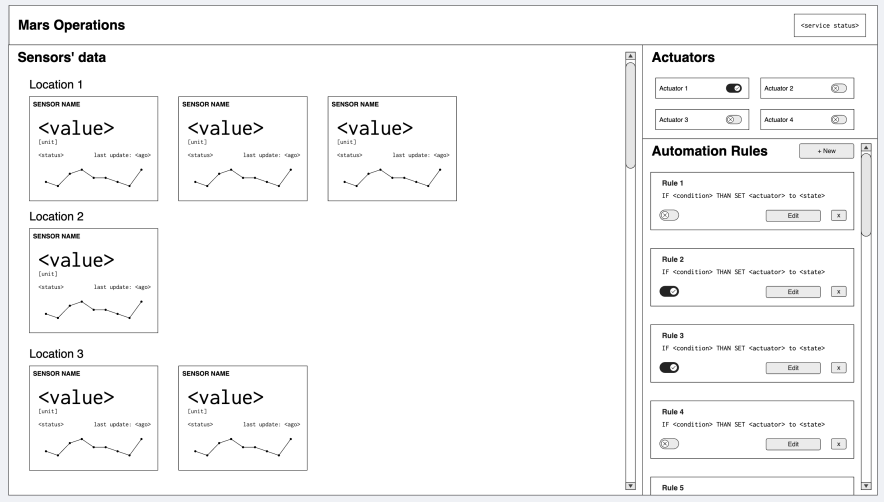
- To ensure safety, the system:
  - **groups** rule outcomes by actuator during a small time window and checks if there some rule that want to set an actuator `ON` and some rule that want to set an actuator `OFF`;
  - chooses a **safe default state** and highlights conflicts in the UI.
- This logic is entirely implemented in the processor-service `Arbitrator` component.

Conflict Handling (Backend)

How conflicts are handled

- When several rules target the same actuator in a short window, the **Arbitrator** groups commands.
- If all rules agree (all `ON` or all `OFF`), the actuator is set accordingly.
- If there was a previous warning for that actuator, it is cleared automatically.
- If at least one rule asks `ON` and at least one asks `OFF`, a **conflict** is detected.
- In conflict, the safe default is applied: actuator set to `OFF`.

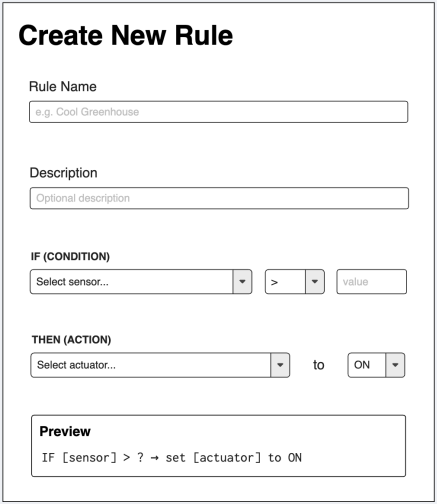
Dashboard Layout (Mockup)



Sensors Panel (Mockup)

- **Grouping by location**
  - Sensors are grouped under headers like *GREENHOUSE*, *ENTRANCE*, *HABITAT*.
  - Each group can display cards in grid or list view.
- **Sensor cards**
  - Show sensor name, location, last value and unit, status indicator and "last update" time.
  - Show a mini SVG chart showing recent history (min/max labels below the chart).
- **Multi-metric sensors**
  - For sensors such as `water_tank_level` or `air_quality_pm25`, the card behaves as a small carousel.
  - Each slide corresponds to a specific metric (e.g. *level pct*, *level liters*, *PM 2.5*).

New Automation Rule Layout (Mockup)



Automation Rules Panel (Mockup)

- **Rule cards**
  - Display the rule name and a readable IF–THEN sentence based on the internal condition.
  - Include a toggle switch to enable/disable the rule (`is_active`).
  - Provide *Edit* and *Delete* buttons; delete is confirmed with a second click.
- **Create / Edit modal**
  - Guides the user through selecting:
    - sensor (and metric, for multi-metric sensors);
    - comparison operator ($<, \leq, ==, >, \geq$);
    - threshold value;
    - target actuator and desired state.
  - Shows a live textual preview of the resulting rule.
- **Conflicts indication**
  - Rules involved in active conflicts are highlighted and display a warning icon with a tooltip.

Actuators Panel (Mockup)

- **Actuator tiles**
  - Each actuator (e.g. *cooling fan*, *habitat heater*) appears as a rectangular tile.
  - Color and icon indicate ON/OFF status at a glance.
- **Interaction**
  - Clicking a tile toggles its state and sends a POST request through the API Gateway to the simulator.
  - WebSocket updates keep tiles synchronized with actual simulator responses.

Technical Architecture Recap

- **Data ingestion**
  - Ingestion-Service polls the simulator, normalizes payloads into `UnifiedEvent`, and publishes them to RabbitMQ.
- **Processing and rules**
  - Processor-Service consumes events, updates the state cache, evaluates rules and arbitrates actuator commands.
  - Rules and their metadata are persisted in SQLite.
- **Exposure to operators**
  - API-Gateway exposes WebSocket and REST APIs to the frontend.
  - Frontend displays live data, histories, rule configuration, conflicts and actuators in a single dashboard.

Example Scenario: Automatic Cooling

1. Simulator reports a high `greenhouse_temperature` reading.
2. Ingestion-Service polls the value, builds a `UnifiedEvent` and publishes it to RabbitMQ.
3. Processor-Service consumes the event; the rule engine finds a rule such as:
   IF greenhouse temperature > threshold THEN set cooling fan to ON.
4. The rule triggers and emits a command to the arbitrator; if no conflicting rule asks for OFF, the arbitrator sets the cooling fan to ON.
5. The actuator command is sent to the simulator and an `actuator_command` event is published.
6. Gateway forwards events via WebSocket; dashboard updates sensors, rule status, and cooling fan tile.

Thank You!

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